

CHAPTER 1

INTRODUCTION

Since last 10 years, there is a huge increase in number of vehicles. But the current transportation infrastructure and car park facilities are insufficient in sustaining the influx of vehicles on the road. Therefore, problems such as traffic congestion and insufficient parking space inevitably crops up. Car parking occupancy detection are of great importance for an effective management of car parking lots. Knowing in real-time availability of free parking spaces and communicating to the users can be of great help in reducing the queues, improve scalability and the time required to find an empty space in a parking lot. Various measures have been taken in the attempts to overcome the traffic problems. Some methods like sensor-based a computer vision classifier are functional nowadays to reduce the problems regarding parking. But these systems are either very costly or less efficient and less robust.

So, for these kinds of issues the researchers have developed some solutions like parking lot recognition techniques by making use of some objects like video cameras and sensors which are used to detect an empty and occupied state of the parking space. By using YOLO (You OnlyLook Once), which is an object detection algorithm based on Convolutional Neural Network (CNN).

The aim of this is to solve the issue of car parking for some extent which is done by taking videos from surveillance cameras and detecting the occupancy of the parking slot whether the parking space is Empty or Occupied. By using this technique, the main parking lot issue, that is availability of the parking spaces in the parking lot is solved, so that the drivers do not waste muchtime in searching for the parking space.

1.1 Object Detection

Object detection is a computer technology that deals with finding instances of semantic items of a specific class (such as individuals, buildings, or cars) in digital photos and videos. It is related to computer vision and image processing. Face detection and pedestrian detection are two well-studied object detection areas. Object detection can be used in a variety of computer vision applications, such as picture retrieval and video surveillance. Predicting the class of one object in an image is an example of image classification.

Identifying the location of one or more items in a picture and drawing a bounding box around their extent is referred to as object localization. Object detection combines these two tasks by locating and classifying one or more items in a picture. When a user or practitioner says "object recognition," they usually mean "object detection." Beginners may find it difficult to discriminate between various computer vision tasks.

With this example, one can discriminate between these three computer vision tasks: Predict the type or class of an object in an image to do image classification. A single-object image, such as a photograph, is used as an input. The result is a class label (e.g. one or more integers that are mapped to class labels). Localization of Objects: This is accomplished by using a bounding box to locate the presence of items in an image and indicate their location.

An image made up of one or more objects, such as a photograph, is used as an input. One or more bounding bears as an output (e.g. defined by a point, width, and height). Object detection is accomplished by locating the existence of items in an image using a bounding box and types or classes of the discovered objects. A photograph, for example, is an input image that consists of one or more objects. One or more bounding boxes (e.g., determined by a point, width, and height), as well as a class label for each bounding box as output.

- **Methods**

Object detection methods are classified as either neural network-based or non-neural approaches. Non-neural approaches require first defining features using one of the methods below, followed by classification using a technique such as support vector machine (SVM). On the other hand, neural approaches, which are often based on convolutional neural networks, are capable of doing end-to-end object detection without specifying characteristics (CNN).

- 1. Non-neural approaches**

- Based on Haar features, Viola-Jones object detection framework
- (SIFT) Scale-invariant feature transform
- (HOG) Histogram of oriented gradients features

- 2. Neural network approaches**

- Region Proposals
- Single Shot Multi-Box Detector (SSD)
- You Only Look Once (YOLO)

- (Refine-Det) Single Shot Refinement Neural Network for Object Detection
- Retina-Net
- Deformable convolutional networks

- **Uses**

It's used for picture annotation, vehicle counting, activity recognition, face detection, face recognition, and video object co-segmentation, among other things. It's also used to track things, such as a ball during a football game, a cricket bat's movement, or a person in a film.

1.2 Computer vision

Computer vision is an interdisciplinary scientific topic that studies how computers can perceive digital images or movies at a high level. From an engineering standpoint, it aims to comprehend and automate operations that the human visual system is capable of. Methods for capturing, processing, analysing, and understanding digital images, as well as the extraction of high-dimensional data from the actual world in order to produce numerical or symbolic information, such as in the form of judgments, are all examples of computer vision tasks.

In this application, understanding refers to the conversion of visual images (retinal input) into world descriptions that make sense to mental processes and can inspire appropriate action. The disentangling of symbolic information from visual data using models built with the help of geometry, physics, statistics, and learning theory can be considered as image comprehension.

The theory behind artificial systems that extract information from images is the subject of computer vision, a scientific area. Video sequences, various camera perspectives, multi-dimensional data from a 3D scanner, or medical scanning data are all examples of picture data. Computer vision is a technological discipline that aims to apply its ideas and models to the development of computer vision systems. Scene reconstruction, object detection, event detection, video tracking, object recognition, 3D pose estimation, learning, indexing, motion estimates, 3D scene modelling, and picture restoration are all sub-domains of computer vision. The automatic extraction, analysis, and understanding of relevant information from a single image or a sequence of images is the subject of computer vision. It entails the creation of a theoretical and algorithmic foundation for autonomous visual

comprehension. Computer vision is a scientific area that studies the theory behind artificial systems that extract information from images. Video sequences various camera perspectives, or multi-dimensional data from a medical scanner are all examples of picture data. Computer vision, as a technological subject, aims to apply its ideas and models to the development of computer vision systems .

1.3 YOLO

The YOLO ("you only look once") algorithm is popular because it achieves high accuracy while running in real-time. In the sense that it only requires one forward propagation pass through the network to create predictions, this method "only glances once at the image. It then outputs recognised items together with bounding boxes after non-max suppression. YOLO employs & single CNN network for both classification and bounding box localization. You Only Get One Look: Object detection that is unified and real-time.

YOLO v1

YOLOv1 is a single-stage object detection model. Object detection is framed as a regression problem to spatially separated bounding boxes and associated class probabilities. A single neural network predicts bounding boxes and class probabilities directly from full images in one evaluation. Since the whole detection pipeline is a single network, it can be optimized end-to-end directly on detection performance.

The network uses features from the entire image to predict each bounding box. It also predicts all bounding boxes across all classes for an image simultaneously. This means the network reasons globally about the full image and all the objects in the image.

YOLO v2

YOLOv2 is a deep learning algorithm that uses a single neural network to simultaneously predict bounding boxes and class probabilities for objects in an image. It has been used for fire detection by training on images and identifying features that are common to fires, such as colour and texture.

YOLOv2 can achieve high accuracy in object detection tasks and can process images in real time. It is also relatively simple to implement. YOLOv2 may struggle with detecting fires in images with complex backgrounds or in lowlight conditions. It may also require large amounts of data for training

YOLO v3

YOLOv3, an enhanced version of YOLOv2 for object detection, introduces improvements for accuracy while maintaining speed. It predicts bounding boxes using anchor box clusters and incorporates logistic regression to assign an objectiveness score for each box. It employs independent logistic classifiers instead of SoftMax for class predictions, using binary cross-entropy loss during training. YOLOv3 predicts boxes at three scales, akin to FPN, utilizing multi-scale feature extraction through a network called DarkNet53, a fusion of DarkNet19 and a novel residual network, optimized for GPU efficiency. It excels in identifying small objects due to multi-scale predictions but shows comparatively lower performance on medium and large objects.

The model's speed and precision trade-off can be adjusted by varying its size. However, its performance diminishes as IoU thresholds increase, and it struggles to align precisely with ground truth. For sonar datasets with limited samples and low signal-to-noise ratios, a refined version named YOLOv3-DPFIN was proposed. It excels in real-time detection of noise-intensive multi-category sonar objects underwater, achieving higher precision and speed than the original YOLOv3 model.

YOLO v4

YOLO (You Only Look Once) v4 is a state-of-the-art object detection model, renowned for its speed and accuracy in real-time object detection tasks. It is part of the YOLO series, which revolutionized object detection by introducing a single-stage object detection architecture. YOLO v4 is an advancement over its predecessors, aiming to improve accuracy while maintaining impressive real-time performance.

Developed by a team of researchers, YOLO v4 integrates various improvements in architecture design, training techniques, and model optimization to enhance object detection capabilities across different domains. It incorporates features like a more robust backbone network, advanced data augmentation strategies, and innovative training methods to improve detection accuracy, especially for small and challenging objects. YOLO v4 maintains the core principles of its predecessors: high-speed inference and accurate bounding box predictions for multiple objects within an image. This model has garnered attention for its ability to handle real-time object detection tasks efficiently, making it a popular choice in applications like autonomous driving, surveillance, robotics, and more.

YOLO v5

YOLO v5 represents the latest evolution in the renowned YOLO (You Only Look Once) series for object detection. Developed by Ultralytics, this iteration focuses on refining and enhancing the core principles of real-time object detection. It introduces a streamlined convolutional neural network architecture that prioritizes speed and accuracy. YOLO v5 offers different model variants—varying in size from YOLOv5s to YOLOv5x—allowing users to choose models based on their specific trade-offs between model size, speed, and accuracy. Notably, its user-friendly nature simplifies the setup, configuration, and training processes, making it accessible for a broad range of applications such as autonomous vehicles, surveillance systems, robotics, and more, where rapid and precise object detection is pivotal. Overall, YOLO v5 is a significant leap forward in object detection, appreciated for its improved performance, efficiency, and ease of use within the computer vision community.

YOLO v6

YOLOv6, an evolution of the popular You Only Look Once (YOLO) object detection algorithm series, builds upon its predecessors' strengths of speed and accuracy. With enhancements in its architecture, potentially leveraging different backbone networks like CSP Darknet or YOLOv5's design, YOLOv6 aims to improve object detection performance while maintaining real-time processing capabilities. It might introduce novel training strategies and optimization techniques to enhance model generalization and adaptability to diverse datasets. YOLOv6 strives to be user-friendly, offering efficient deployment options for applications like real-time object detection in videos, surveillance systems, autonomous vehicles, and more. For the most recent and detailed information about YOLOv6, consulting the latest research publications and official resources is recommended.

YOLO v7

YOLO (You Only Look Once) v7 stands as the seventh iteration of the groundbreaking object detection algorithm. Renowned for its real-time detection capabilities, YOLO revolutionized the field by processing images in a single pass through the neural network, swiftly identifying multiple objects within a single frame. This approach, dividing the image into a grid and predicting bounding boxes and class probabilities for each grid cell, allows for simultaneous object detection. YOLO v7 maintains the balance between accuracy and speed, leveraging advanced backbone architectures for feature extraction, thereby enhancing its ability to

comprehend complex visual patterns. Continuously evolving, each version of YOLO aims to refine object detection performance by integrating novel techniques and architectural improvements into its framework.

YOLO v8

YOLO v8, the latest iteration of the YOLO (You Only Look Once) series. YOLOv8 represents a significant milestone in computer vision, designed to enhance the efficiency and precision of object detection. Its underlying principle simplifies the process by processing an entire image through a neural network known as a convolutional neural network (CNN) in a single pass, eliminating the need for laboriously inspecting each part of the image individually. This streamlined approach, often referred to as “You Only Look Once,” leads to impressive gains in both speed and efficiency.

- **Principle followed by YOLO v8**

YOLOv8 follows a straightforward set of principles. It all starts with an input image, which can be of varying sizes, making it a versatile tool for a wide range of applications. The image then undergoes a process of feature extraction within a neural network. This network, carefully engineered and trained, excels at identifying and highlighting the most relevant features within the image, akin to a detective piecing together crucial clues at a crime scene. YOLOv8 stands out by not just detecting objects at a single scale but by simultaneously spotting them at different scales, ensuring it can identify objects both large and small effectively.

- **The Role of Anchor Boxes in YOLO v8**

Key to YOLOv8’s prowess in object detection is its utilization of anchor boxes. These anchor boxes act as predefined templates for various objects, aiding the algorithm in making informed predictions about object locations and sizes within specific grid cells. By leveraging anchor boxes, YOLOv8 enhances its accuracy and precision, further bolstering its ability to recognize objects in images accurately. It’s akin to having a set of standardized models that help YOLOv8 determine the shape and size of the objects it encounters, contributing to more accurate predictions.

- **YOLOv8’s Advancements Over Previous Versions**

When we compare YOLOv8 with its predecessors, it becomes evident that it brings significant advancements to the table. It combines the speed and efficiency that were

hallmarks of earlier YOLO versions with improved accuracy and adaptability. YOLOv8's ability to handle various input sizes and its array of specialized variants cater to a broad spectrum of real-world applications, from real-time tasks on resource-constrained devices to challenging scenarios involving blurred images or intricate datasets. Its enhanced generalization, fewer false positives, and improved non-maximum suppression techniques further underline its reliability in practical use cases. As computer vision continues to evolve, YOLOv8 serves as a testament to the ongoing quest for more refined, faster, and more precise object detection algorithms, with the potential to revolutionize industries such as autonomous vehicles, healthcare, and beyond. In conclusion, YOLOv8 is a monumental step forward in the realm of object detection, symbolizing the relentless pursuit of better solutions in the exciting field of computer vision.

1.4 Organization of Report

The report consists of 5 chapters. In chapter 1, the preview of car parking slot detection and introduction to object detection, computer vision, YOLO algorithm. Problem statement and the objectives of the car parking slot detection using YOLO v8 is discussed in chapter 2, a literature survey on project is discussed in chapter 3, system design is explained in chapter 4, the conclusion is given in the last chapter.

CHAPTER 2

PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

As the population grows, there is an increase in the private vehicles which caused many issues to the people at the crowded areas such as shopping malls, restaurants and universities parking lots. This results in time waste for the drivers in search for the empty parking space which includes atmospheric pollution and frustration of the drivers. Developing a project that uses a camera to record real-time data and feed it to a system to discover vacant car parking slots in a parking area. Car parking slot detection using YOLO algorithm which calculates the number of vacant parking slots in a parking area based on the system's confidence level.

2.2 Objectives

- To capture images from web camera.
- To process the captured images.
- To identify vacant and occupied slots.
- To colour vacant and occupied slot with green and red respectively.
- To display number of vacant slots.

CHAPTER 3

LITERATURE SURVEY

Literature survey is the important part of report as it gives direction in the area of research. It helps to set goals for the analysis, thus giving problem statement. It is also nematic and thorough search of all types of published literature as well as other sources. Following are some of the research papers related to this project.

Bandi Sairam and Gopi Krishna [1] from the department of Information Technology National Institute of Technology, Raipur, India, presented research on **Automated Vehicle Parking Slot Detection System Using Deep Learning**. This method focuses on the use of Mask R CNN architecture for solving instance segmentation problems in deep learning. Specifically, it discusses how the model is used to detect objects in images, provide their bounding boxes and classes, and create a mask around the objects. Additionally, the use of PSDL (Parking Slot Detection System) for detecting marking points, determining the type of parking slot, and identifying free space between vehicles using sensors. This combination of technologies aims to address automated vehicle parking slot detection.

The implementations discussed revolve around utilizing vision-based car parking technology, particularly in the context of detecting parking slots using web cameras. It emphasizes the challenges and cost considerations associated with sensor-based slot detection systems and highlights the preparation and maintenance of infrastructure as a major concern.

The conclusion focuses on the successful implementation of the Mask R CNN method, achieving significant accuracy in recognizing object masks and detecting object bounding boxes. Additionally, it emphasizes the simulation-based nature of the method used, as well as the cost considerations and infrastructure maintenance concerns associated with sensor-based slot detection systems. The approach specifically focuses on vision-based car parking, leveraging web cameras for slot detection, and provides insights into the accuracy and practical implications of the proposed method.

Raj Patel and Praveen Meduri [2] from the department of Computer Engineering, California State University Sacramento, USA, presented research on **Car Detection Based Algorithm for Automatic Parking Space Detection**. This method focuses on a car

detection-based algorithm for automatic parking space mapping and detection. It discusses an approach that aims to achieve a high level of automation in mapping parking lots, with a particular emphasis on detecting parked cars and available parking spaces. The approach is designed to have its highest performance over a busy parking lot during the busiest time, indicating its practical application in real-world scenarios.

The methodology presented in the method focuses on an approach for automatic parking space and car detection. It takes advantage of the fact that parked cars tend to remain stationary for longer periods. The approach is designed to significantly reduce the human effort involved in mapping parking lots, with results indicating a compelling 90% reduction. One of the strengths of this approach is that the recall of the parking space detection scheme increases with the number of parked cars, and in theory, it could approach 100% detection on a day when all parking spots are fully occupied. It also emphasizes the generalizability of the approach across different parking lots and camera angles, attributing its success to its focus on car features rather than parking lot features.

The implementation details specifies where the performance of the approach is evaluated. It discusses the selection of threshold values for parameters such as the IoU and the frame rate, which are crucial in determining the accuracy of the car tracking approach. Notably, it highlights that while these hyperparameters require tuning, they only marginally reduce accuracy when within a reasonable range.

Thus, it concludes the method's ability to effectively identify a significant number of parking spaces is highlighted. It emphasizes the approach's potential to greatly reduce the human effort involved in mapping parking lots, leading to a compelling 90% reduction. Moreover, the paper discusses the theoretical capability of the approach to achieve close to 100% detection of all parking spots on days when the parking lot is fully occupied. It emphasized the approach's consistent performance across different parking lots and camera angles, attributing this to the focus on car features rather than parking lot features. The ability of the approach to detect parking spaces even with faint parking lines is also underscored.

Moran JU and Haibo LUO [3] from the department of Institutes for Robotics and Intelligent Manufacturing, Chinese, presented research on **A Simple and Efficient Network for Small Target Detection**. This uses “dilated module” to expand the receptive field without loss of resolution or coverage. Then the applied feature fusion in different dilated modules to improve the ability of the network in detecting small targets. Finally, used

“passthrough module” to get the finer-grained information from the earlier layer and combined it with the semantic information from the deeper layer. To improve the detection speed of the network, it uses 1×1 convolution to reduce the dimension of the network. The composed small vehicle dataset based on VEDAI dataset and DOTA dataset, respectively, and also analysed the distribution of the small targets in each dataset. To evaluate the performance of the network, trained the model on the dataset above and compared with the state-of-the-art target detection algorithms. The method utilizes several key techniques to improve small target detection. The use of dilated modules to expand the receptive field, passthrough modules to combine fine-grained and semantic information, and feature fusion all contribute to enhancing the network's ability to detect small targets. Additionally, the method incorporates 1×1 convolutions to reduce the dimension of the network, resulting in reduced computation and improved detection speed.

The conclusion emphasizes a simple, fast, and efficient network designed for small target detection. The network's effectiveness is highlighted through quantitative and qualitative comparisons with state-of-the-art networks. It is noted that the method improves the ability to detect small targets and accurately locate them. The authors detail the use of dilated modules to expand the receptive field, the implementation of feature fusion, and the utilization of passthrough modules to enhance the network's performance.

Hanbo Zhou^a and Wei Xiang^c [4] from the department of Southwest Minzu University, China, presented research on **Method for judging parking status based on yolov2 target detection algorithm**. The method focuses on based machine vision to distinguish parking spaces. This method uses the yolov2 target detection algorithm, uses the camera to collect the image information of the parking spaces and vehicles, uses the yolov2 target detection algorithm to build the vehicle and parking space contour detection model, and obtains the parking space and vehicle center coordinate information, and calculates the vehicle and parking space contours through the parking space discrimination algorithm To determine whether there is a vehicle parked in the parking space, determine the status of the parking space (occupied/vacant).The experiment outlines a methodology for judging parking status based on the YOLOv2 target detection algorithm. The method based on machine vision to distinguish parking spaces and effectively utilize public parking space resources on the roadside, aiming to realize smart transportation more comprehensively.

Afterwards, the experiment was conducted. Firstly, they collected more than 600 pictures of moving and stationary vehicles during both day and night using a drone. These images were then used to establish a vehicle contour recognition model. The dataset was divided into a training set and a test set in an 8:2 ratio. To enhance the precision and accuracy of the dataset, data augmentation techniques were employed. This involved flipping the pictures, zooming in on the partial dataset, and simulating various complex environments. The experiment involved data collection, dataset division, data augmentation, and the application of the YOLOv2 algorithm for parking space identification and vehicle center coordinate extraction, ultimately leading to the determination of parking space status.

Aakash K. Shetty and Yashkumar J. Patel [5] from the department of Computer Engineering, NMIMS University Mumbai, India, presented research on **A Review: Object Detection Models**. This method focuses on comparative analysis of different object detection techniques such as R-CNN, Fast R-CNN, Faster R-CNN, YOLOv1, YOLOv2, YOLOv3, YOLOv4, and SSD. It addresses the challenges in comparing these methods, highlighting the trade-off between precision and speed for real-life applications. Furthermore, discusses the use of data augmentation to increase image variability, thereby enhancing the robustness of the models in various environments. It also touches upon optimizing the training method to achieve greater precision without increasing the cost of inference.

The methodology involves a comparative analysis of different object detection techniques, including two-stage models such as R-CNN, Fast R-CNN, and Faster R-CNN, as well as single-stage models like YOLOv1, YOLOv2, YOLOv3, YOLOv4, and SSD. The comparison focuses on parameters such as precision, backbone, accuracy, frames per second (FPS) provided, and computational time. It deduces various conclusions based on the results obtained from the comparative analysis of different object detection techniques. It emphasizes that each algorithm is equally good for its application, and the selection of a specific algorithm should align with the application model. Furthermore, It highlights the importance of exploring existing techniques, performing comparative analyses, and suggesting potential implementations for optimal results in different sectors. It also indicates that the experimental results showed single-stage object detection techniques to be more optimal compared to two-stage object detection techniques.

Cui Gao and Qiang Cai [6] from National Engineering Laboratory for Agri-product Quality Traceability, Beijing Technology and Business University Beijing Key Laboratory of

Big Data Technology, BTBU Beijing Technology and Business University Beijing, China, presents research on **YOLOv4 Object Detection Algorithm with Efficient Channel Attention Mechanism**. This method focuses on various advanced techniques and algorithms related to object detection, attention mechanisms, and deep learning. Specifically, it discusses object detection using YOLOv4 target detection algorithm, attention mechanism, and the application of these techniques in the field of deep learning for areas such as image recognition and processing. It seems to delve into the intricacies of these methods and their significance in computer vision and artificial intelligence.

The comprehensive methodology used in method, likely in the context of evaluating the effectiveness of object detection and attention mechanisms in deep learning. It mentions the use of the YOLOv4 target detection algorithm, attention mechanisms, and the measurement of various parameters such as mAP (mean Average Precision) and detection speed. The methodology also appears to involve a comparative analysis with other works, potentially aiming to highlight the significance of the techniques.

It concludes that, an improved YOLOv4 object detection algorithm, attention-YOLOv4, with the efficient channel Attention mechanism. This algorithm can improve the accuracy at a relatively low additional computational cost. The algorithm introduced channel attention mechanism ECANet into the residual module of YOLOv4's backbone network, CSPDarknet53, and added different weights to the channel features to make the network pay more attention to the important features containing the object to be detected, while suppressing the invalid features affecting the final detection results, thus enhancing the robustness of the algorithm. Through experimental comparison, the detection accuracy of this algorithm has been improved to some extent.

Se-Ho Park and Hyoseok Yoon [7] from the department of Korea Electronics Technology Institute, Mapo-gu, Seoul, Korea, presented research on **An All-in-One Vehicle Type and License Plate Recognition System Using YOLOv4**. The development of an innovative system that addresses Korean License Plate Recognition (LPR) with vehicle type detection, showcasing its potential for various complex smart city applications. It also highlights the involvement of multiple authors in the conceptualization, methodology, software, validation, and other aspects of the research.

The methodology utilized involves an in-depth approach to address Korean License Plate Recognition (LPR) with vehicle type detection. The authors' contributions include

various aspects such as conceptualization, methodology, software, validation, formal analysis, data curation, as well as writing and preparation of the original draft. The research also involved the development of an integrated embedded platform and a commercial prototype equipped with high-speed LPR function, demonstrating the comprehensive nature of the study.

The implementation involves the use of YOLOv4 as the underlying object detector to realize the proposed Korean License Plate Recognition (LPR) system with vehicle type detection. The researchers utilized an open source darknet framework to train YOLOv4 for the detection of custom classes including vehicle types, license plates, and license plate characters. This approach demonstrates the utilization of deep learning technology to address the identification of various targets in smart surveillance and urban mobility applications.

Jingyi Zhao, Zhanlin Ji (Member, IEEE) and Ivan Ganchev (Senior Member, IEEE) [8] from department of Telecommunications Research Centre (TRC), University of Limerick, V94 T9PX Limerick, Ireland presented research on **Improved Vision-Based Vehicle Detection and Classification by Optimized YOLOv4**. In this method, an attention mechanism is utilized to suppress the interference features of images through both channel dimension and spatial dimension. In addition, a modification of the Feature Pyramid Network (FPN) part of the Path Aggregation Network (PAN), utilized by YOLOv4, is applied in order to enhance further the effective features through down-sampling. This way, the objects can be steadily positioned in the 3D space and the object detection and classification performance of the model can be improved.

Object detection and classification are crucial for intelligent transportation systems and various other applications. Existing object detection methods can be categorized as hardware-based and vision-based methods. Traditional feature-based object detection methods have been replaced by deep learning (DL) feature-based methods. Current DL feature-based methods are considered state of the art due to their object detection accuracy and operational speed. Many object detection models are based on CNNs due to their rich representation power. Recent advances in object detection have been driven by the success of R CNNs and other incarnations. The YOLOv4_AF model utilizes an attention mechanism to suppress interference features of images and enhances effective features through down sampling. Experiments demonstrate that YOLOv4_AF outperforms the original YOLOv4 model and other state-of-the-art models in terms of mean average precision (mAP) and F1

score. In this, a more precise vehicle detection and classification model, based on YOLOv4 with applied additional optimization, has been presented. The main idea, incorporated into it, was to increase the receptive field in both channel and spatial dimensions by introducing an attention mechanism in the form of a CBAM module.

Guan Sheng Wong and Aznul Qalid Md. Sabri [9] from department of Computer Science and Information Technology, University of Malaya, presented research paper on **Review of Vision-Based Deep Learning Parking Slot Detection on Surround View Images**. They explore the vision-based deep learning methods for parking slot detection. They categorize these methods into four main categories: object detection, image segmentation, regression, and graph neural network, and provide detailed explanations and insights into the unique features and strengths of each category. Additionally, they analyse the performance of these methods using three widely used datasets: the Tongji Parking-slot Dataset 2.0 (ps 2.0), Sejong National University (SNU) dataset, and panoramic surround view (PSV) dataset, which have played a crucial role in assessing advancements in parking slot detection.

The datasets mentioned include PS, PSV, and SNU with details on resolution, camera view, and the number of images for training, validation, and testing. Additionally, the document highlights the ps 2.0 dataset as widely used in the field of parking slot detection, featuring a diverse range of parking lot environments and three distinct parking lot types. The document highlights the utilization of various datasets for evaluation and talks about the methods used to generalize effectively across different datasets. Additionally, it references the exploration of techniques to reduce the model's FLOPS during inference and leveraging embedded systems specifically designed for running deep learning models. The findings in the document offer valuable insights for researchers and practitioners in this field, guiding future research directions and advancements in vision-based parking slot detection using deep learning techniques.

The conclusion, provides a comprehensive overview of vision-based deep learning approaches for parking slot detection on surround view images. It specifically focuses on deep learning methods and categorizes them into four main categories: object detection, image segmentation, regression, and graph neural network. Each category is thoroughly explained and reviewed, highlighting the key features and specialties of the methods.

Upendra Suddamalla and Balaji Ravichandiran [10] from MooVita Pte. Ltd., Singapore presented research on **Camera Based Parking Slot Detection For Autonomous Parking**. This method using a novel combination of computer vision techniques on the camera data mounted on sides of a car. Autonomous driving and driver assistance systems are emerging areas to enhance human safety and driving experience. Autonomous parking requires understanding the parking slot position, dimension, and occupancy status, achieved through various sensors such as Lidars, ultrasonic range detectors, radars, and cameras. The Existing techniques using sensors have limitations in providing accurate markings and boundaries of parking slots, particularly when no reference object is available. Camera-based methods have been proposed but face challenges with varying parking conditions, illumination, and noise. The method addresses the need for a generic approach to handle dynamic parking environments and random obstacles.

The conclusion focuses the effectiveness of the algorithm in addressing the problem of free parking slot detection. The algorithm is capable of detecting both cars and trees present in the input image, demonstrating its robustness in real-world scenarios. Additionally, it is noted that there is no standard public dataset available for the problem of interest, emphasizing the significance of the dataset collected for the research. The complete system of the algorithm has been depicted as a flow diagram to illustrate its functionality.

Fatma M. Talaat and Hanaa Zain Eldin [11] from the department of Computers and Control Systems Engineering, Faculty of Engineering, Mansoura University, Egypt, presented research on “**An improved fire detection approach based on YOLO-v8 for smart cities**”. This improved fire detection approach for smart cities based on the YOLOv8 algorithm, called the smart fire detection system (SFDS), which leverages the strengths of deep learning to detect fire-specific features in real time. The SFDS approach has the potential to improve the accuracy of fire detection, reduce false alarms, and be cost-effective compared to traditional fire detection methods. It can also be extended to detect other objects of interest in smart cities, such as gas leaks or flooding. The framework for a smart city consists of four primary layers: Application layer, Fog layer, Cloud layer, and IoT layer. The algorithm utilizes Fog and Cloud computing, along with the IoT layer, to collect and process data in real time, enabling faster response times and reducing the risk of damage to property and human life.

The conclusions are evident that the system has made a valuable contribution to the field of computer vision and real-time fire detection. The system also seems to have potential for further improvement and application in real-world scenarios. The possibility of using OCNN and correlation methods in the future work to achieve better results. Additionally, the document emphasizes the importance of a secure and open access infrastructure, user-centered architecture, and the capability to store, retrieve, share, and access a vast amount of public and private data.

Nabin Sharma and Rathachai Chawuthai [12] from Department of Computer Engineering, School of Engineering, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand, presented research on **Parking Time Violation Tracking Using YOLOv8 and Tracking Algorithms**. This presents an approach that can introduce a low-cost time violation tracking system using CCTV, Deep Learning models, and object tracking algorithms. This approach is fairly new because of its appliance of the SOTA detection technique, object tracking approach, and time boundary implementations. YOLOv8, along with the Deep SORT/OC-SORT algorithm, is utilized for the detection and tracking that allows us to set a timer and track the time violation. Using the same apparatus along with Deep Learning models and algorithms has produced a better system with better performance. Also introduces a real-time parking time violation tracking algorithm using closed-circuit cameras and DL models with a tracking algorithm to persist the information from one frame to its subsequent frame. The algorithm utilizes a state-of-the-art object detection algorithm called YOLOv8 for detecting vehicles. For tracking vehicles, two object tracking algorithms were implemented, Deep SORT (Simple Online Real Tracking) and Observation Centric SORT (OC-SORT).

The conclusion mainly highlights the effectiveness and significance of the algorithm for parking violation detection in a time series context, citing its accurate and reliable solution to the parking time violation problem. It acknowledges the innovative approach, emphasizing the exploration of state-of-the-art object detection and tracking models, and the novel frame-by-frame evaluation with respect to time. Additionally, the conclusion includes a review of related work, details of the parking time violation process, and the presentation of results.

Xiangwu Ding and Ruidi Yang [13] from Donghua University, shanghai, shanghai, China presented research on **Vehicle and Parking Space Detection Based on Improved**

YOLO Network Model. This is based on YOLO v3 network and applied to parking spaces and vehicle detection in parking lots. Based on YOLO v3, this adds a residual structure to extract deep vehicle parking space features, and uses four different scale feature maps for object detection, so that deep networks can extract more fine-grained features. Experiment results show that this method can improve the detection accuracy of vehicle and parking space, while reducing the missed detection rate. The current research dataset comes from three datasets, the PASCAL VOC dataset, the COCO dataset, and the PKLot dataset. The PASCAL VOC and COCO data sets are common data sets for object detection, with 20 categories and 80 categories respectively. This experiment only uses the three data sets car, bus, and truck. The PKLot dataset provides an image dataset for parking space classification. The PKLot dataset is not the format of the PASCAL VOC dataset.

The improved YOLO v3 algorithm model is based on the feature pyramid network structure, and the feature maps of different levels are merged and connected to obtain four sets of predicted feature maps, and the position and class predictions are performed on the four sets of predicted feature maps. This mainly studies the problem of parking space detection. The vehicle and parking space detection method based on the improved YOLO v3 algorithm. The improved algorithm is applied to vehicle and parking space detection and it has achieved good results.

Daniel Padilla Carrasco and Domènec Puig [14] from department of Electronic and Communications Technology, Universidad Autónoma de Madrid, Spain, presented research on **T-YOLO: Tiny Vehicle Detection Based on YOLO and Multi-Scale Convolutional Neural Networks**. In this, the model can detect large, small, and tiny objects. Specifically, the use of a multi-scale mechanism to learn deep discriminative feature representations at different scales and automatically determine the most suitable scales for detecting objects in a scene (i.e., in our case vehicles). The multi-scale module reduces the number of trainable parameters compared to the original YOLO-v5 architecture. The experimental results also demonstrate that precision is improved by a large margin. The PKLot1 dataset, structured based on parking locations, meteorological states, and specific days, contains approximately 4474 images with almost 100 tagged parking spots. These images include the location of bounding boxes and occupancy, totalling around 424,269 tagged spots. The dataset was split into 80% for training and 20% for testing. Additionally, data augmentation techniques were employed using YOLOv5 standard augmentation, incorporating mosaic/mix-up, letterbox, perspective, HSV colour space, and flipping. The model was implemented using a system

comprising a GPU of Nvidia 2080 Ti and the PyTorch library. The YOLOv5 baseline was used, employing hyperparameters and configurations consistent with YOLOv5. The model was validated on the PKLot dataset, utilizing both a COCO-trained YOLOv5 model and a PKLot fine-tuned YOLOv5 model. It also mentions the introduction of a new layer in YOLOv5 that resulted in a significant outperformance of the baseline model. Additionally, the impact of spatial and channel information using attention modules on the performance of tiny object detection is discussed.

Mamoona Humayun and Marwah Khalid Alsadun [15] from the department of Information Systems and Department of Computer Science, College of Computer and Information Sciences, Jouf University, Saudi Arabia, presented research on **Traffic Management: Multi-Scale Vehicle Detection in Varying Weather Conditions Using YOLOv4 and Spatial Pyramid Pooling Network**. This focuses on the development and application of a methodology for object detection in traffic scenarios using the DAWN dataset, a collection of 1000 photos from actual traffic scenarios. The data annotation process, which involves labeling images in YOLO_Darknet format using a free and open-source tool called Label Image. They also mention the necessity of adaptive data fusion in infrastructure augmented autonomous driving systems in a brief. This suggests a focus on advanced technologies for autonomous driving systems.

The methodology used involves the development and application of a methodology for object detection in traffic scenarios using the DAWN dataset, which consists of 1000 photos from actual traffic scenarios. The researchers discuss the data annotation process, utilizing the YOLO_Darknet format and the open-source tool LabelImg for labeling images. Additionally, it mentions an extension of the vehicle detection algorithm for vehicle trajectory analysis, where vehicle location, speed, acceleration, and jerk are used as time functions to detect normal and abnormal vehicle motion patterns. Furthermore, it includes image augmentation techniques related to specific weather conditions.

The implementation details are focused on the use of the DAWN dataset, which contains 1000 photos from real traffic scenarios, to train the model. The data annotation process involved labeling images in YOLO_Darknet format using the open-source tool LabelImg. Additionally, it discusses an extension of the vehicle detection algorithm for vehicle trajectory analysis, utilizing vehicle location, speed, acceleration, and jerk as time functions to detect normal and abnormal vehicle motion patterns using a video dataset.

CHAPTER 4

SYSTEM DESIGN

System design is the conceptual design that defines the structure and behaviour of a system. A design description is a formal description of a system, organized in a way that supports reasoning about the structural properties of the system. It defines the system components or building blocks and provides a plan from which products can be procured, and systems developed, that will work together to implement the overall system.

4.1 System Architecture

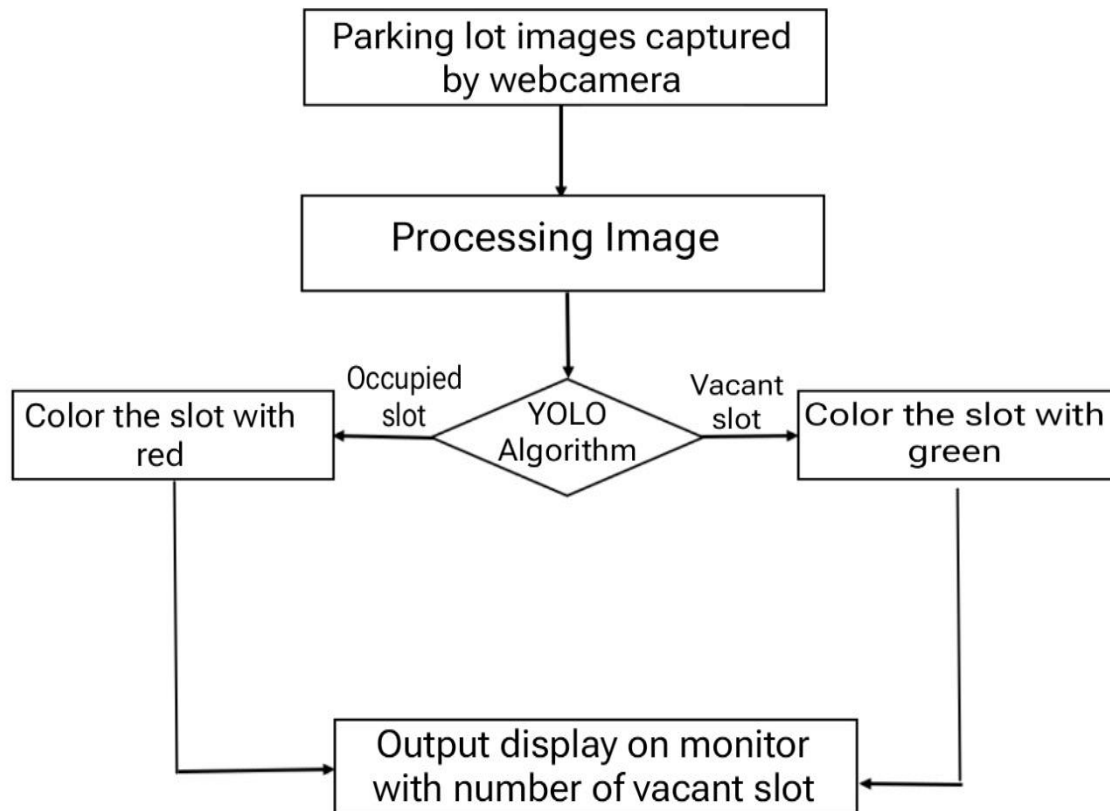


Figure 4.1: System Architecture

4.2 Description

4.2.1 Input Data

Image or video feed from web cameras placed in the parking lot serves as input data for the system. These cameras capture real-time footage of the parking area.

4.2.2 Preprocessing Image

The input data might require preprocessing, such as resizing, normalization, or enhancement, to improve the quality of the images or video frames. This step ensures that the input is suitable for the YOLO algorithm.

4.2.3 YOLO Algorithm

YOLO is an object detection algorithm that divides an image into a grid and predicts bounding boxes and class probabilities for each grid cell. YOLO can identify and locate multiple objects within an image. In the case of parking slot detection, it's trained to recognize empty and occupied parking spaces.

- **Object Detection**

YOLO processes the input images or video frames and detects parking slots by drawing bounding boxes around the identified slots. The algorithm also assigns number of slots being empty.

- **Decision Making**

Once the YOLO algorithm identifies parking slots, the vacant slots bounding box coloured in green and the occupied slot are coloured in red.

4.2.4 Output

The monitor screen will present a live feed of the parking lot captured by cameras, with vacant and occupied parking slots visually indicated through color-coded. Vacant slots might be highlighted in green, while occupied ones could be marked in red, allowing for quick identification of available spaces. Simultaneously, a numerical count of available parking slots will be displayed as cars enter or leave the area. This user-friendly interface provides an instant overview of the parking lot's occupancy status, enabling users to swiftly locate and navigate to empty parking spaces.

Overall, the architecture for car parking slot detection using the YOLO algorithm involves a pipeline that takes input from cameras, processes it using the YOLO model, detects parking slots, and provides actionable information based on the detected data.

CHAPTER 5

CONCLUSION

One of the most common problems is to find parking slots, which has become a serious issue in contemporary urban mobility as the number of vehicles are increasing day by day. The traditional system of finding parking slots manually is quite challenging in practical usage. Based on the literature survey, the implementation of the YOLO (You Only Look Once) algorithm for car parking lot detection showcases its effectiveness in accurately identifying and delineating parking spaces. Research papers consistently demonstrate YOLO's capability to efficiently detect vehicles within parking lots, aiding in occupancy detection and management. The algorithm's real-time processing and high accuracy make it a promising solution for automating parking space monitoring systems, contributing significantly to smart city initiatives and traffic management.

REFERENCES

- [1] Bandi Sairam and Gopi Krishna "Automated Vehicle Parking Slot Detection System Using Deep Learning " Proceedings of the Fourth International Conference on Computing Methodologies and Communication (ICCMC 2020) IEEE Xplore Part Number: CFP20K25-ART; ISBN:978-1-7281-4889-2
- [2] Raj Patel and Praveen Meduri "Car Detection Based Algorithm for Automatic Parking Space Detection" 2020 19th IEEE International Conference on Machine Learning and Applications (ICMLA)
- [3] Moran JU and Haibo LUO "A Simple and Efficient Network for Small Target Detection" 2020 19th IEEE International Conference on Machine Learning and Applications (ICMLA)
- [4] Hanbo Zhoua and Wei Xiangc "Method for judging parking status based on yolov2 target detection algorithm" The 8th International Conference on Information Technology and Quantitative Management (ITQM 2020 & 2021)
- [5] Aakash K. Shetty and Yashkumar J. Patel "A Review: Object Detection Models" 2021 6th International Conference for Convergence in Technology (I2CT) Pune, India. Apr 02-04, 2021
- [6] Cui Gao and Qiang Cai "YOLOv4 Object Detection Algorithm with Efficient Channel Attention Mechanism" 2020 5th International Conference on Mechanical, Control and Computer Engineering (ICMCCE)
- [7] Se-Ho Park and Hyoseok Yoon "An All-in-One Vehicle Type and License Plate Recognition System Using YOLOv4" 2022 by the authors. Licensee MDPI
- [8] Jingyi Zhao, Zhanlin Ji (Member, IEEE) and Ivan Ganchev (Senior Member, IEEE) "Improved Vision-Based Vehicle Detection and Classification by Optimized YOLOv4" 2021 19th IEEE International Conference on Machine Learning and Applications (ICMLA)
- [9] Guan Sheng Wong , Kah Ong Michael Goh and Aznul Qalid Md. Sabri "Review of Vision-Based Deep Learning Parking Slot Detection on Surround View Images" 2023 by the authors. Licensee MDPI

- [10] Upendra Suddamalla and Balaji Ravichandiran “Camera Based Parking Slot Detection For Autonomous Parking” (2020).
- [11] Fatma M. Talaat and Hanaa Zain Eldin “An improved fire detection approach based on YOLO-v8 for smart cities” Neural Computing and Applications (2023)
- [12] Nabin Sharma and Rathachai Chawuthai “Parking Time Violation Tracking Using YOLOv8 and Tracking Algorithms” 2023 by the authors. Licensee MDPI
- [13] Xiangwu Ding and Ruidi Yang “Vehicle and Parking Space Detection Based on Improved YOLO Network Model” ICAITA 2019
- [14] Daniel Padilla Carrasco and Domènec Puig “T-YOLO: Tiny Vehicle Detection Based on YOLO and Multi-Scale Convolutional Neural Networks” 2021 19th IEEE International Conference on Machine Learning and Applications (ICMLA)
- [15] Mamoon Humayun and Marwah Khalid Alsadun " Traffic Management: Multi-Scale Vehicle Detection in Varying Weather Conditions Using YOLOv4 and Spatial Pyramid Pooling Network " 2022 by the authors. Licensee MDPI